ENGINEERING PROBLEM / DISTRIBUTION GRID RESILIENCE

EV CHARGING LOAD IMPACT SIMULATOR

Impact Modeling, Thermal Degradation Analysis, and Microgrid Mitigation for High-Penetration EV Charging | Standards: IEEE C57.91 • ANSI C84.1

01 / PROBLEM CONTEXT & INFRASTRUCTURE VULNERABILITY

Urban and suburban MV/LV distribution grids were designed for passive, unidirectional power flows and predictable demand diversity. Uncoordinated EV charging—Level 2 (7.2–19.2 kW) and DC fast charging (50–150 kW)—during the 17:00–21:00 residential arrival window stacks new demand on top of the legacy evening peak. In the representative 2.5 MVA / 12.47 kV feeder scenario, 70% EV penetration raises peak demand from **1.52 MW to 4.46 MW**, producing an **87.7% overload**. The feeder voltage falls to **0.951 p.u.**, while transformer winding hot-spot temperature reaches **189.4°C**. IEEE C57.91 indicates a **74.2× aging acceleration factor**; static TOU schedules also create a synchronized 23:00 rebound peak of **4.31 MW**.

02 / PROJECT OBJECTIVES & ENGINEERING SCOPE

01 STOCHASTIC FLEET CHARACTERIZATION
Model 3,500+ empirical EV sessions: arrival times, dwell durations, 40–98 kWh battery sizes, and flexibility index $\Phi = t_{dwell} / t_{min}$.

02 COUPLED POWER FLOW & THERMAL AGING
Build a 13-bus radial LinDistFlow voltage solver and dynamic top-oil / hot-spot heat-transfer equations aligned with IEEE C57.91.

03 SUSTAINABLE MICROGRID & V2G OPTIMIZATION
Coordinate smart valley-filling with an 800 kWp solar PV array, 1.2 MWh BESS, and bidirectional Vehicle-to-Grid (V2G) peak shaving.

04 INTERACTIVE DECISION-SUPPORT PLATFORM
Deliver a full-stack, real-time simulator to stress-test EV penetration scenarios and quantify asset degradation and carbon abatement.

03 / QUANTITATIVE ENGINEERING BENCHMARKS & MITIGATION TARGETS

PERFORMANCE METRIC	UNCONTROLLED LEGACY	STATIC TOU REBOUND	VOLTGRID™ MICROGRID	ENGINEERING STANDARD
Peak feeder demand	4.46 MW (+87.7%)	4.31 MW (+81.6%)	**2.45 MW (+3.2%)**	≤ 2.50 MW (rated)
Maximum hot-spot temperature	189.4 °C	176.2 °C	**97.0 °C**	< 110.0 °C (IEEE C57.91)
Aging factor (F_EQ)	74.19×	24.11×	**0.095× (preserved)**	≤ 1.0× (design base)
Minimum bus voltage	0.951 p.u.	0.953 p.u.	**0.973 p.u.**	≥ 0.950 p.u. (ANSI C84.1)
Daily CO2 emissions	14,868 kg	11,764 kg	**11,206 kg (–24.6%)**	Maximize solar self-supply

MITIGATION GOAL	Keep peak demand near rated capacity	Protect transformer thermal life	Improve voltage & reduce emissions